



**An Explainable AI Framework for Identity Fraud Detection at Bank Account Opening
Using Stacking Ensemble Learning and SHAP Analysis**

By

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Abstract

Identifying fraudulent account openings is challenging due to the absence of transaction histories in applications, which are essential for most fraud detection systems, and the dynamic nature of the data. We examine interpretable machine learning methodologies utilising the BAF NeurIPS 2022 dataset (1,000,000 applications, 1.10% fraudulent cases). We assess a stacking ensemble comprising XGBoost, LightGBM, and Logistic Regression in comparison to five key benchmark models, alongside a four-learner Stack+Random Forest variation as an ablation, while maintaining the remainder of the architecture unchanged. Training was done during months 0–5, threshold calibration during month 6 and testing during month 7. Models were evaluated based on TPR@5%FPR, PR-AUC, ROC-AUC, and frozen-threshold metrics, employing stratified and paired bootstrap resampling to measure uncertainty. Kernel SHAP elucidated the framework, augmented by feature elimination, cross-architecture concordance, and seed consistency.

In the seventh month, the stack achieved its peak TPR@5%FPR (0.5770) and ROC-AUC (0.8970), whereas LightGBM attained the optimal PR-AUC (0.2172). At its frozen threshold, the stack attained a recall of 0.5217 with a false-positive rate of 0.0391. The stack was distinguishable from XGBoost concerning TPR@5%FPR and PR-AUC, although not from LightGBM. A fairness assessment conducted in the seventh month revealed a predictive-equality rate ratio of 0.2884 across different age demographics. In a controlled cross-variant shift, TPR@5%FPR decreased to 0.4644. The framework demonstrated robust predictive capabilities and valuable interpretability, although offered few advantages from increased complexity through stacking. The findings suggest a necessity for prudence about fairness, the dependability of explanations, and oversight during distribution shifts.

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Chapter 1: Introduction

1.1 Background

Identifying fraudulent bank account openings is structurally different from identifying fraudulent bank account transactions, because, at the time of opening a bank account, neither the account holder nor the bank has a history of transactions. Furthermore, the account opening processes capture varying identities, financial data, device data, and behavioural data. Therefore, to identify fraudulent bank account openings, we need to leverage data from the account opening process. As described by Jesus et al. (2022), the Bank Account Fraud (BAF) dataset captures the challenges surrounding fraudulent bank account openings, including high imbalance, temporal challenges, and intentional bias.

Fraudulent applications can include attributes that appear individually valid and, therefore, can be difficult to detect unless the relationships between attributes are considered. Khan et al. (2023) and Chang, Shih and Lin (2023) highlight the importance of relationships between identity attributes and between account identity, respectively, for detecting synthetic identities and fraud.

Non-linear relationships can be captured using boosting algorithms, and stacked generalization leverages a meta-model to combine multiple learners (Wolpert, 1992; Chen and Guestrin, 2016; Ke et al. (2017). However, by increasing the complexity of the structure, we negatively affect the easiness of interpreting the model. A more complex model only justifies an increase in complexity if it provides sufficient improvement in predictability.

Incomplete automated systems lead to false positives which negatively affect legitimate applicants by requiring an investigation which delays onboarding or leads to an unwarranted rejection. Therefore, automated systems must consider both fraud detection and be subjected to case scrutiny. The Information Commissioner's Office and The Alan Turing Institute (2022) stress that responsible use of AI to aid decision-making includes the provision of an adequate explanation. Furthermore, Regulation (EU) 2024/1689 sets further requirements in the areas of transparency and risk management, but retains systems used for the detection of financial fraud outside the field of creditworthiness of high risk. As such, explainability is addressed in the context of responsible model governance rather than as evidence of compliance with legal requirements.

1.2 Problem Statement

SHapley Additive exPlanations (SHAP) assign local and global attributions to features for machine learning predictions (Lundberg and Lee, 2017). SHAP offers attributions, but we can't say attributions are reliable. SHAP doesn't show that a feature that is important to fraud is causally related to fraud, that fraud prediction depends on the feature, that the same ranking would be produced in different architectures, or that explanations would be the same after retraining. Kumar et al. (2020) point out limitations when interpreting Shapley values as feature importance metrics, while Aas, Jullum and Loland (2021) show that feature dependence can impact explanations. Rudin (2019) argues against relying on post-hoc explanations if simple models can be created that are interpretable and perform similarly.

Evaluating the systems is also difficult because fraud data are generally highly imbalanced. Although accuracy can be high, the system may not be good at detecting fraud, potentially warranting a precision-

recall based evaluation (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Moreover, numerical gains in performance metrics do not necessarily justify a more complex model, as the models may be fit too closely to the data.

1.3 Research Gap and Contribution

Recent BAF research has closed some of the gaps. Sun et al. (2025) studied the impact of architecture and operating points on BAF subject to an extreme imbalance of class distributions. Bhuiyan et al. (2026) studied the explanation reliability across BAF variants and found an explanation-reliability gap, within which stable SHAP rankings and performance-based feature reliance do not always align. This paper does not claim to be the first investigation of explanation reliability in the context of BAF. Instead, this paper studies a heterogeneous stacking ensemble which combines XGBoost, LightGBM and Logistic Regression, with Logistic Regression as the meta-learner. The ensemble is then interpreted using model-agnostic SHAP and compared with individual models and simpler baselines.

The combined approach offers four novel features in one integrated framework: assessment of the temporal impact on performance at a fixed false-positive-rate operating point; SHAP reliability assessment via fidelity, cross-architecture alignment, and stability; a paired bootstrap analysis of whether added complexity provides architectural performance gains; and testing of subgroup fairness and cross-variant robustness with BAF Variant IV, which is assessed at a fixed threshold. This study, therefore, examines tradeoffs and considers the impact of complexity on performance, explanation reliability, and robustness, as opposed to assuming that a more complex approach is better.

1.4 Aims and Objectives

The aim is to build an explainable AI system to identify identity fraud when opening a bank account using the BAF NeurIPS 2022 dataset and to evaluate the quality of the provided explanations to determine if they would withstand interrogation of individual decisions.

The objectives are:

- To critically evaluate the research on explainable AI combined with ensemble learning and identity fraud detection.
- To define BAF and describe important characteristics of identity fraud.
- To build and evaluate a stacking ensemble model versus other individual models and benchmarking models.
- To determine the deployed model through SHAP and examine fidelity, agreement, and stability.
- To assess model performance in the presence of severe imbalance, high uncertainty, disparities in different subgroups, and changes in the distribution of data.

1.5 Research Questions

- **RQ1:** To what extent can a stacking ensemble detect identity fraud when opening a bank account under a distribution shift along the temporal dimension, and at a fixed operating point?

- **RQ2:** To what extent does SHAP adequately explain the deployed model, is the explanation sufficient to ensure substantive scrutiny of individual adverse decisions for which the model has been built?
- **RQ3:** Does stacking gradient boosting models and gradient boosting over an interpretable linear model get sufficient benefits to offset the loss of interpretability?

Chapter 2: Literature Review

2.1 Scope and Organisation of the Review

In six stages, this chapter establishes the research context, machine-learning fraud detection and its orientation toward transaction rather than application fraud; severe class imbalance and appropriate evaluation practice; ensemble learning and the conditions under which stacking provides gains; explainability and the methods responsible for its production; the distinction between generating explanations and establishing their reliability; and fairness and governance. It concludes by synthesising these strands to resolve the research gap that this study has identified.

2.2 From Transaction Fraud to Bank-Account-Opening Fraud

From expert-based rules and statistics to machine learning based detection systems, financial fraud detection has made substantial progress. Statistical frameworks for fraud detection have been studied in Bolton and Hand (2002), while Ngai et al., (2011), Abdallah, Maarof and Zainal (2016) and West and Bhattacharya (2016) categorize the next phase of fraud detection as supervised and unsupervised learning. While machine learning techniques can learn to identify fraudulent behaviour, rule-based systems can easily be understood due to their dependency on predefined rules and will remain efficient. Many empirical studies examine credit card and payment transactions, where different methods have been developed based on the accessibility of transaction records. In particular, Whitrow et al. (2009) develop summary features based on recent transactions. Bahnsen et al. (2016) build on this work, and Jurgovsky et al. (2018) cast the problem of fraud detection as a sequence classification problem. Bhattacharyya et al. (2011) review data mining techniques. Carcillo et al. (2018, 2021) study streaming detection and the integration of supervised and unsupervised learning.

These techniques are significantly related to the account history. At the account opening stage, there is no history. New applicants are classified as customers. Therefore, during the account opening stage, decisions are made based on identity, financial, device and session information that are collected during the customer journey. For this reason, techniques such as transaction aggregation, behavioural profiling and sequence modeling cannot be applied, and as a result, the studies on transactional fraud detection cannot be generalized to the application stage. Studies specific to identity and account opening fraud are also scarce. For example, Khan et al. (2023) use inductive graph convolutional networks to recognise synthetic identities.

Conversely, Chang, Shih, and Lin (2023) present an uncertain graph framework that simulates the connections between identity and account data. Both suggest that combinations of identification traits may result in fraud signals. Jesus et al. (2022) address the shortage of public benchmarks with the Bank Account Fraud (BAF) suite. BAF has many millions of records and examples of severe class imbalance, and temporal bias. BAF has intentional bias and is completely synthetic, so it permits privacy preserving research. That said, when considering controlled benchmark, the performance scores of BAF should be treated as controlled benchmark evidence. BAF has also been designed so that some variables have been defined as negative sentinel values to represent missing information. Whether these patterns are predictive is an empirical question. Therefore, predictive modeling using BAF is described in Chapter 3, as opposed to contemporary fraud modeling techniques.

2.3 Severe Class Imbalance and the Evaluation Problem

Fraud detection is a well-known example of an imbalanced-classification problem (He and Garcia, 2009; Branco, Torgo and Ribeiro, 2016). In such cases, fraud accounts for only a small percentage of cases, and a classifier may learn to predict the 'legitimate' class with high accuracy. For such cases, accuracy is inadequate as a measure of fraud-detection ability. Two broad approaches to tackling imbalance exist. The first is data-level approaches which focus on shifting the distribution in training, with SMOTE used to create synthetic minority examples through interpolation (Chawla et al., 2002). The fraud literature, such as Talukder, Khalid, and Uddin (2024), continues to use these strategies. The cost-sensitive method of Bahnsen, Aouada, and Ottersten (2014) is an example of the algorithm-level technique, which focuses on changing learning by class or mistake weighting. Albalawi and Dardouri (2025) investigate the same subject, emphasising both conventional methods and deep learning.

There are methodological differences between the two approaches. First, undersampling can change predicted probabilities and therefore, requires great care for calibration. Although Kabane (2024) is a preprint and therefore, should be cautiously examined, it illustrates the leakage risk on resampling before data partitioning. Cost-sensitive learning will avoid creating synthetic observations, whereas class weighting will not guarantee calibrated probabilities. This distinction will be important for the setting where thresholds on held out data will be chosen and left unchanged for succeeding observations. The evaluation metrics will also have to consider the problem of the minority class. The common treatment of the ROC analysis is given by Fawcett (2006). Meanwhile, Davis and Goadrich (2006) describe the relation between ROC and precision-recall assessment.

In very extreme imbalance, ROC analysis may show good performance as the false positive is normalised to a big negative class. In this situation, Saito and Rehmsmeier (2015) argue that precision-recall analysis becomes the more insightful way of analysis. Jesus et al. (2022) combine aggregate evaluation with an explicit operating point by measuring the true-positive rate at a 5% false-positive rate. This operating point balances the trade-off between falsely flagging legitimate applicants and detecting fraudulent applications. The consideration at the same false-positive rate enables the comparison of models.

Evaluation over time is an even more significant concern. The population of applicants and fraud patterns may change over time. Therefore, performance estimated by random partitioning may be overstated when observations used for building the model are similar to future observations. Unlike random partitioning, partitioning over the BAF monthly structure allows a realistic evaluation of temporal patterns in fraud. Several studies have documented the consequences of the design of BAF evaluations. For example, Sun et al. (2025) evaluated several models including Logistic Regression, radial-basis Support Vector Machine, Random Forest, LightGBM, and Gated Recurrent Unit, and observed significant differences in fraud detection depending on the learning goal and operating conditions. Based on these observations, Sun et al. (2025) concluded that the choice of operating point and objective function has a significant influence on the design of fraud detection systems. This contradicts the belief that, without a doubt, highly sophisticated models of fraud detection will excel over less complex models.

2.4 Ensemble Learning and the Conditions for Stacking Gains

Generalization is improved with ensemble methods by combining multiple learners. Random Forest defines an Average over bootstraps, while gradient boosting methods define Average over iterations. Aggregation is generally performed by XGBoost and LightGBM for structured tabular data, which define an additional gradient boosting methods. Stacked generalization follows a more distinct approach. Instead of averaging in a predefined way, Wolpert (1992) suggested using a meta-learner to combine the predictions of several base learners. During the development of the meta-model, out-of-fold predictions are used so that the base learners are responsible for generating predictions for observations not used to develop the base learners, and the meta-learner combines these predictions.

The basis of stacking is the diversity among constituent models. When predictions are highly correlated, components add little new and distinct information to the meta-model. This is particularly evident with multiple successive gradient boosted models defined on the same predictors. Therefore, stacking may not lead to an improvement over its strongest constituent; its value can only be shown empirically. Financial-fraud studies report promising ensemble results, and there have been a number of recent studies in this area. For instance, Talukder, Khalid and Uddin (2024) use a multistage ensemble approach for transaction fraud. Meanwhile, Vijayanand and Smrithy (2025) integrate ensemble learning and explainable AI for mobile-money fraud detection. Almalki and Masud (2025) examine stacking and explainability. Velarde et al. (2023) study gradient boosting in the context of both balanced and imbalanced datasets. Both studies are preprints, and so are used here as supplementary rather than primary sources.

A common issue with studies that probe models in comparison is their reliance on point estimates. A more complex model may receive a higher score. However, this difference may be small and are within sampling variations. For this reason, judgments of model superiority need to consider ranking and sampling uncertainty. In the case of comparable performance and the additional burden of explainable stacking, the tradeoff of additional complexity and opacity may not be justified.

2.5 Explainability and SHAP

Increasing predictive complexity leads to an increase in the difficulty of model understanding. Guidotti et al (2018) examine the literature on explaining black-box models and Arrieta et al. (2020) consider explainability as part of the responsible AI. However, interpretability cannot be viewed as an absolute concept. Lipton (2018) offers the distinction of model transparency as an inherent property of a model versus transparency as a post-hoc explanatory property. Doshi-Velez and Kim (2017) state that interpretability should not simply be assumed or treated as a given, but should be assessed. Post-hoc interpretability is dominated by LIME and SHAP. LIME uses an interpretable local model to explain individual predictions (Ribeiro, Singh and Guestrin, 2016). SHAP uses Shapley values to create additive feature importance values (Lundberg and Lee, 2017). This model also explains global features and their values by cumulative SHAP values at different locations.

Tree ensembles also have a model-specific post-hoc interpretability approach, TreeSHAP (Lundberg et al., 2020). However, a heterogeneous stacking architecture that integrates the tree and linear models cannot be interpreted as a single tree ensemble for TreeSHAP. If one focuses on the prediction function of the deployed integrated model as a whole, rather than the base learners, this calls for a model-

agnostic SHAP approach. SHAP is finding new applications in financial fraud research. Lin and Gao (2022) create group SHAP for financial fraud interpretation. Zhou et al. (2023) explore user-centred explainability and Psychoula et al. (2021) study explainable machine learning for fraud detection.

There is a significant critical literature that argues generating attribution is not the same as verifying it. Kumar et al. (2020) describe situations where Shapley values should not be interpreted as feature importance when interpreted directly. Aas, Jullum and Loland (2021) show that predictor dependence can influence attribution in a significant manner. Additionally, SHAP does not provide a mechanism for establishing a causal relationship between a certain characteristic and fraud.

Rudin (2019) asks the challenging question in a high-stakes situation, should the gap between the predictive performance of two models be significant enough for opaquely explaining one model and the other model be interpretable if they provide the same predictive performance? This question is particularly important for this study, as the two aspects of this evaluation have to be balanced.

2.6 From Explanation to Explanation Reliability

The previous literature claims that explanation generation on its own is not a reliable measure. Regarding reliability, there is no established approach that universal consensus agrees upon for evaluating a set of SHAP attributions. Following Doshi-Velez and Kim (2017), who state interpretability should be assessed empirically, this study examines three complementary empirically assessable properties: predictive relevance, cross-architecture agreement, and stability.

The first of these is assessed via a fidelity-based feature-removal check. Evidence is provided for the information importance of those variables if removing features ranked highly by SHAP adversely affects predictive performance. Such removal should be interpreted with great care, especially if the predictors are correlated. It indicates the importance of predictive information and is not to be taken as a statement of importance or to be taken as a universal definition of explanation fidelity. The second of these is cross-architecture agreement. In the case of models of different architectures displaying similar ranking by a global measure, the absence of such a ranking in other architectures suggests that the ranking is not unique to that architecture. It should be noted that agreement is not the same as correctness. Different models can yield the same ranking of variables even though they are spurious.

The third property is stability. Architectures which behave differently when retrained under different random initializations yield different global rankings, and, consequently, are harder to consider reproducible. Conversely, stable rankings give us confidence in how well architectures hold up to the perturbations studied, and while stability may suggest that an explanation is likely to be the correct one, there are other considerations. Bhuiyan et al. (2026) describe the most relevant empirical evidence. Using the BAF suite, they contrast SHAP stability and permutation-based importance in the context of feature relevance under domain shift for Logistic Regression, Linear Support Vector Classification, and Random Forest as well as other gradient-boosted models. They observe a degree of explainability-reliability trade-off where SHAP rankings are comparatively stable while output-based relevance for some higher ranked features is either lower or more variable.

An important detail is that SHAP and permutation measures assess different parameters. Thus, we should not view their conflicting views as an explanatory failure. On the contrary, Bhuiyan et al. (2026) cite that stability of explanation does not guarantee stability of predictive dependence.

2.7 Fairness and Responsible AI Governance

Errors in fraud detection can impact applicant populations differently. Jesus et al. (2022) use predictive equality based on group false positive rates because false positives unfairly flag legitimate applicants. They indicate age, income and employment status as the protected attributes under BAF, whereas their benchmark fairness analysis mainly uses age as a proxy. Fairness, of course, is not absolute. Conflating false positive rates with fraud detection rates and group size could mean that no one number will account for the disparity. Therefore, fairness measures should account for size, fraud rates and error rates.

The focus on fairness within responsible AI promotes this approach. The Information Commissioner's Office and The Alan Turing Institute (2022) focus on the need for substantive explanations for AI-based decisions, while the NIST AI Risk Management Framework sees explainability and transparency as part of the risk management of trustworthy AI (Tabassi, 2023). Regulation (EU) 2024/1689 offers a broader, risk-based framework for governance that includes the transparency of systems, data governance and human oversight of systems in the regulated sectors (European Union, 2024). However, systems for financial fraud detection are excluded from the high-risk category of creditworthiness in Annex III. So, responsible governance based on documented scrutiny can be justified within the current legal context; but, adopting SHAP to portray a fraud detector should not be interpreted to mean that the legal requirements have been fulfilled.

2.8 Synthesis and Research Gap

Four conclusions have emerged from the literature. First, when compared to the financial fraud detection research, the research on account opening detection lacks the behavioural profiles that commonly characterize established feature engineering approaches. Second, when compared to one another, both accuracy-focused approaches and the prospective evaluation of fraud detection systems are subject to severe class imbalance and frequent changes that make them inadequate for realistic comparisons. Third, when compared to one another, the benefit from stacking depends largely on the complementarity of the models, and greater complexity of the model should not be expected to provide greater benefits. Fourth, SHAP does provide attributions, but internal consistency or stable rank orders do not provide evidence of either predictive reliability, causal importance, or explanation relevance.

Recently, BAF research has closed much of the remaining gap. Sun et al. (2025) showed that extreme class imbalances demonstrate the importance of architecture, operating points, and choices of objectives, of which, at least, two (architecture and objective) are interchangeable. Bhuiyan et al. (2026) identified a gap between stable SHAP attribution and performance-based feature relevance. Consequently, this study will not claim that the BAF research ignored the comparison of models and the reliability of explanations in bank account opening fraud detection. This review of the literature seeks to fill the joint evaluation of architectural complexity and explanation reliability of a heterogeneous stacking model in bank account opening fraud detection. Within the studies of BAF reviewed, explanation validation was more focused on the different individual model architectures compared to the stacking model of a linear and gradient boosting model. Additionally, studies of BAF lacked the joint analysis focusing on the explanation of the model and the integration of the architectural complexity.

This study fills this gap by evaluating a heterogeneous XGBoost–LightGBM–Logistic Regression stack under temporal partitioning and a matched operating point, statistically assessing its performance relative to simpler alternatives, and evaluating its SHAP explanations through fidelity-oriented feature

removal, cross-architecture agreement and stability. With this framework, the study is able to determine whether an ensemble performed better than what could be achieved by simpler alternatives and if such ensemble can be explained. The ideas in this approach are based on the gap described above and this forms the basis of the Chapter 3 methods.

Chapter 3: Methodology

3.1 Research Design

This study uses a quantitative experiment on existing data with no primary or human data collected. Competing models are created with the same constraints. They are then assessed with techniques that are appropriate for severe class imbalance. Afterward, each model goes through explainability, fairness, and robustness testing. The design is more of a comparison design, not an exploratory design: each of the candidates is trained on the same data, tuned under the same search procedure, calibrated to the same operating constraint, and is evaluated on data that was set aside and not used in the model development. RQ3 requires this, since the only way to assess if increasing complexity is justified at the cost of decreasing interpretability, is to evaluate each model in the same way. Figure 1 provides the details of how each dataset is used.

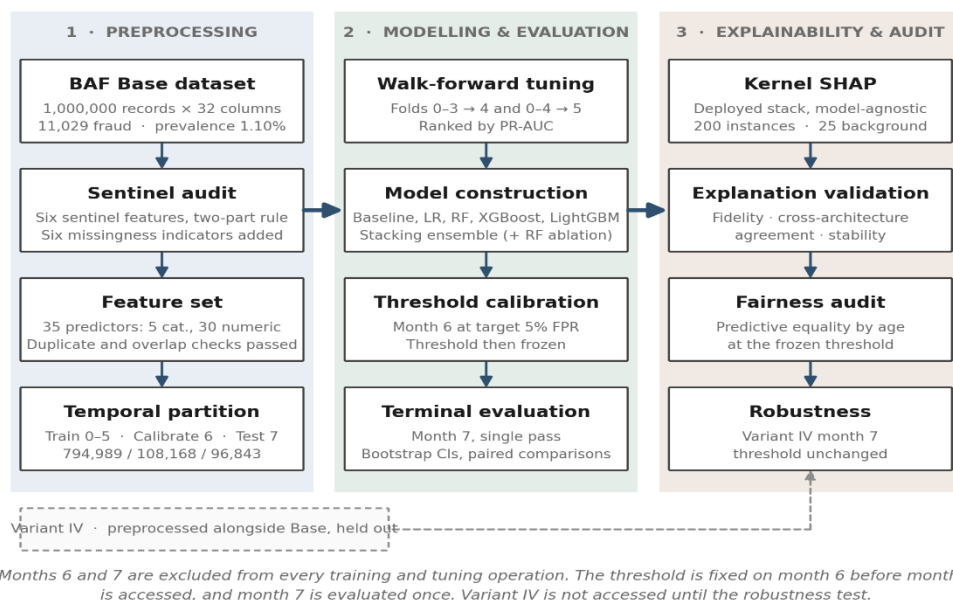


Figure 1. End-to-End Experimental Framework. Source: Author's construction based on the implemented experimental workflow.

3.2 Dataset

The study uses a suite of fraud cases known as Bank Account Fraud (BAF) developed from a real-world, anonymized dataset of opening bank accounts by Jesus et al. (2022). The dataset underwent privacy-

preserving transformations before being trained with a conditional generative adversarial network. Instances were checked for source dataset matching to further reduce risk of disclosure.

Two datasets were used. Base is used for model development and primary evaluation, while Variant IV is used for a controlled cross-variant distribution-shift stress test. Variant IV introduces a controlled temporal discrepancy in prevalence disparities between protected groups and uses the same overall schema and generative framework as Base (Jesus et al., 2022) and therefore is not considered for independent external validation.

Variant IV was processed during the data preparation stage so that the same transformations could be made. However, it was not used for model development, hyperparameter selection or threshold adjustment. The last month of Variant IV was used for the final robustness analysis. Each dataset has 1,000,000 records. There are 32 original columns containing 30 predictive features, a binary target of `fraud_bool` and a month column with a span of 8 months. For Base, 11,029 applications were considered fraudulent with a prevalence of around 1.10%. A scoping qualification is necessary. Jesus et al. (2022) described the application fraud domain as containing frauds that are perpetrated by real-world identity impersonation or the construction of fake identities. The BAF label does not distinguish these mechanisms. Therefore, in this study, the claims are focused at the application-stage identity fraud rather than at the synthetic identity fraud.

The data used in the study is synthetic and available to the public, so the study is not required to process identifiable banking-customer data.

3.3 Data Characterisation and Preprocessing

3.3.1 Sentinel-Encoded Missingness

Standard null-value inspection shows no conventional missing values, since BAF assigns negative values to missing values throughout many variables. An audit of the six documented fields showed approximately 0.04% to 74.25% of encoded missingness. The fields `intended_balcon_amount` and `prev_address_months_count` showed the largest percentage of missingness, followed by `bank_months_count`.

Since negative values do not always indicate missing values, the numerical variables were analyzed individually. Five fields, `prev_address_months_count`, `current_address_months_count`, `session_length_in_minutes`, `bank_months_count`, and `device_distinct_emails_8w`, contained the repeated sentinel value `-1`. Negative values for variables such as `credit_risk_score` and `velocity_6h` were considered valid and retained, as they contained many valid negative values instead of a single missingness code. `intended_balcon_amount` required special treatment. As documented in BAF, negative values in this field are to be considered missing, and therefore not following the single value `-1` pattern, negative values in `intended_balcon_amount` were treated as missing.

3.3.2 Missingness as a Predictive Signal

Rather than imputing the six sentinel-coded fields, their original values were retained and a binary missingness indicator was created for each. The target-oriented descriptive analysis was confined to the training period (months 0–5) to ensure that validation and test results were excluded from the

evaluation of missingness patterns. During this training period, the absence of `prev_address_months_count` correlated with a fraud rate of 0.0132, in contrast to 0.0028 when the value was available, resulting in a ratio of roughly 4.73. The relevant ratios were roughly 2.58 for `intended_balcon_amount` and 1.80 for `bank_months_count`. These results represent descriptive correlations and should not be construed as proof that absence leads to fraud. These findings were strictly treated as associations and not evidence that missingness causes fraud or that any mechanism of identity fraud explains the relationship. It should also be noted that the descriptive analysis was not used to exclude unfavorable indicators. This was done by including all six pre-defined indicators of missingness in modeling. The same transformations were applied to Base and Variant IV.

3.3.3 Duplicates and Leakage Checks

No duplicate redundancies were found in any dataset, no duplicate predictor vectors were present within Base, or no predictor-vector overlap was found in the first 5 months of development vs. the 7th month test. These checks lessen the possibility that final performance is overstated due to repeated data.

3.3.4 Feature Set and Encoding

Three fields were not used for modeling. The target for prediction was `fraud_bool`. The field `month` was used only for creating temporal segments. The field `device_fraud_count` was eliminated as it was degenerate in the training data with a single value observed and zero variance. Once the six missingness fields were added, the final predictor set consist of 35 fields: five categorical and 30 numerical fields. Two preprocessing pipelines were constructed. One-hot encoding was performed for categorical fields and unevaluated categories at prediction time were ignored. Numerical fields were unaltered for tree-based models. The Logistic Regression pipeline performed numerical fields standardization with `StandardScaler`. All learned transformations were performed within the model pipelines using relevant training data.

3.4 Temporal Partitioning

In Base, the prevalence of fraud varied in the eight months. Fraud levels were 1.47% in month seven and 0.87% in month two. The application volumes also exhibited variation. Due to these factors, a sequential as opposed to a random order for evaluation of models was justified.

Table 1. Temporal partition of the BAF

Partition	Months	Applications	Fraud cases	Fraud prevalence
Training	0–5	794,989	8,151	1.03%
Validation	6	108,168	1,450	1.34%
Test	7	96,843	1,428	1.47%
Total	0–7	1,000,000	11,029	1.10%

The training window spans months 0-5, where the calibration month is reserved for month 6. Month 7 comprises the terminal evaluation. Jesus et al. (2022) train for 6 months and test for the last two

months, and urge the adoption of this split in other scenarios, particularly in cases where the dataset is reserved for validation of hyperparameters or threshold definition. In line with the recommendation, the three-way partition is also adopted in this work. Month 7 was not involved in any of: model training, hyperparameter tuning, or calibration. Observations from the testing period were made in the contexts of descriptive data quality and temporal audits. These observations were not used in the fitting of models or in the selection of modeling strategies.

Fraud cases were shown to increase across the three partitions of the development data, suggesting temporal changes in the distribution of the class. Hence, the three-way partition is used to distinguish model training from threshold selection and performance evaluation. Although this design sacrifices some of the fraud cases available for the terminal evaluation compared to keeping months 6 and 7 together, it provides a validation period for threshold refinement and a subsequent period for final evaluation.

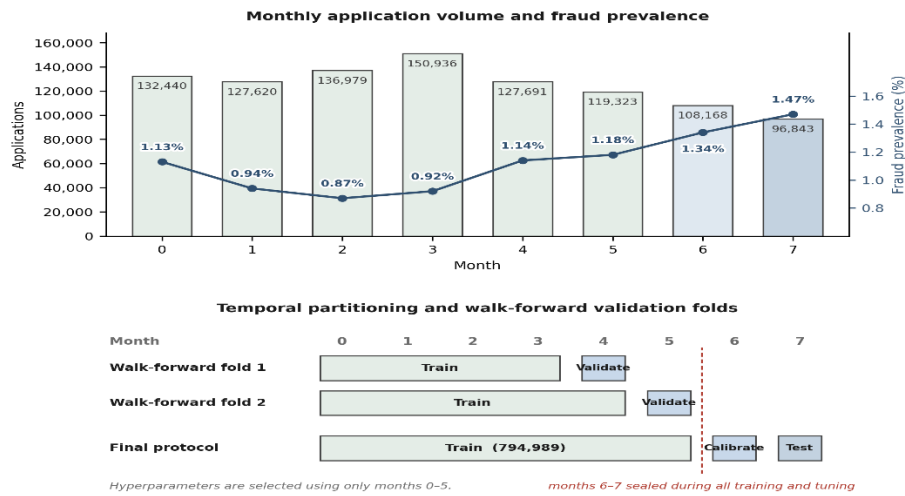


Figure 2. Temporal Partitioning and Walk-Forward Model Development.

3.5 Treatment of Class Imbalance

Instead of SMOTE or other forms of synthetic oversampling, class imbalance was addressed using algorithmic-level weighting (Chawla et al., 2002). Using this method prevents a system from creating any new minority class observations and prevents the specific leakage that can occur when resampling is done incorrectly before partitioning, which Kabane (2024) describes for the context of gradient-boosted fraud detection. Dal Pozzolo et al. (2015) demonstrate how SMOTE and similar techniques can under or overestimate probability distributions, which can impact decision thresholds. The weighting methodology used in this paper is similar to Bahnsen, Aouada and Ottersten (2014) where the impact of imbalanced decisions in credit-based scenarios can be mitigated without manipulating the original dataset.

Logistic Regression, Random Forest and LightGBM used balanced class weighting. XGBoost used `scale_pos_weight`, set as the ratio of legitimate to fraudulent training observations. For the final months 0–5 training set, this ratio was approximately 96.53. During each round of walk-forward hyperparameter tuning, XGBoost class ratio was recalculated for each fold of the training dataset separately, ensuring that the validation month had no influence on class weight.

3.6 Model Construction and Hyperparameter Selection

The model comparison included a majority-class Dummy Classifier, Logistic Regression, Random Forest, XGBoost, LightGBM and heterogeneous stacking ensemble. The Dummy Classifier establishes a no-skill baseline. Logistic Regression provides a somewhat easier to understand linear benchmark as Rudin (2019) argues that in high-stakes cases interpretable models often perform as well as black-box models and any claim regarding improved performance for a more complex architecture should be evaluated against Logistic Regression. Random Forest is a bagging-based ensemble (Breiman, 2001) while XGBoost and LightGBM provide gradient-boosted tree implementations (Friedman, 2001; Chen and Guestrin, 2016; Ke et al., 2017).

Hyper-parameters were chosen using walk-forward validation constrained to the months 0–5 training window:

- Fold 1: months 0–3 predicting month 4.
- Fold 2: months 0–4 predicting month 5.

Candidate configurations were ranked based on PR-AUC.

Table 2. Selected model configurations

Model	Search space	Selected configuration
Logistic Regression	$C \in \{0.01, 0.1, 1.0, 10.0\}$	$C = 0.01$, balanced class weighting
Random Forest	Estimators $\{200, 300\}$; maximum depth $\{8, 16\}$; minimum leaf size $\{5, 20\}$	200 estimators, maximum depth 16, minimum leaf size 20
XGBoost	Estimators $\{300, 500\}$; maximum depth $\{4, 6\}$; learning rate $\{0.05, 0.1\}$	500 estimators, maximum depth 4, learning rate 0.05
LightGBM	Estimators $\{300, 500\}$; leaves $\{15, 31\}$; learning rate $\{0.05, 0.1\}$	300 estimators, 15 leaves, learning rate 0.05

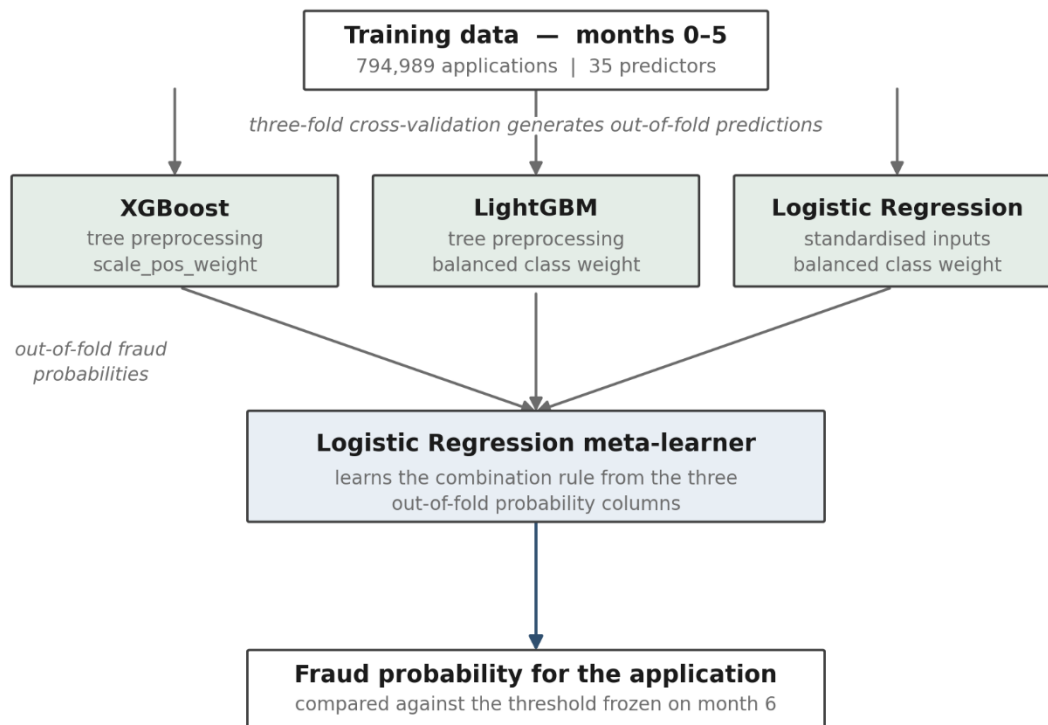
The search spaces were deliberately restricted to make computations manageable. The configurations selected should be considered as the best configured settings within the tested regions rather than optimal hyperparameters. For Logistic Regression, the tested C value resulted in mean walk-forward PR-AUC values that were identical to four places after the decimal, therefore, the selected value should not be interpreted as evidencing a significant superior value for that degree of regularization.

3.6.1 Stacking Ensemble

Stacks composed of XGBoost, LightGBM and Logistic Regression, with the final meta-learner being Logistic Regression, are constructed following the stacked generalization principle of Wolpert (1992). Each base-learner retains its specific preprocessing pipeline, and predicted values are provided to the meta-model as probabilities. The implementation is based on scikit-learn's StackingClassifier with `stack_method="predict_proba"` and three-fold cross-validation ($cv=3$) to generate out-of-fold base predictions to train the meta-learner.

The internal $cv=3$ procedure implements standard stratified, non-temporal, out-of-fold folds without shuffling. While all observations in these folds are exclusively in the months 0–5 training window, these

folds do not preserve temporal order. Thus, observations from later training months may be used to build a base learner to make out-of-fold predictions for training month observations. All validation month 6 observations and test month 7 observations are completely excluded from this internal stacking procedure. Thus, this is a methodological limitation when compared to a fully temporally ordered out-of-fold stacking methodology, but no validation- nor test-set leakage is introduced.



The four-learner variant adding Random Forest is evaluated as an ablation under the identical protocol.

Figure 3. Primary Heterogenous Stacking

3.7. Evaluation Protocol

Each model undergoes the same steps for outer evaluation: train from months 0 to 5, select a threshold on month 6 and keep it frozen during the evaluation on month 7. For each model, the validation ROC curve is used, and the threshold that is closest to a 5% false positive rate is selected. Jesus et al., (2022) use this as the operating point to balance the detection of fraudulent applications and the burden caused to legitimate applicants due to false positives.

Two different types of evaluation metrics are reported, the first type of metrics are TPR@5%FPR. Each model's test-set ROC curve is interpolated to obtain the value at FPR=0.05, giving the measurement of the ranking performance at the common operating point of the ROC curve. The second type of evaluation metrics are precision, recall, F1-score, accuracy, balanced accuracy and achieved FPR, and are calculated using the threshold that was selected in the validation month 6. These metrics describe the effect of crossing the frozen threshold from the validation month 6 to the remaining test months. For this reason, TPR@5%FPR and the metrics of frozen-threshold recall are not comparable. TPR@5%FPR is obtained from the test ROC curve at exactly 5% FPR, whereas frozen-threshold recall is the sensitivity of the validation selected threshold when applied to month 7 without further adjustment.

In addition, threshold-free ranking metrics of PR-AUC and ROC-AUC are reported, where the calculation of ROC-AUC follows the standard practice of ROC analysis as described by Fawcett (2006). PR-AUC is highlighted due to the nature of severe class imbalance (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015) and accuracy is reported, in the context of severe class imbalance, only to show why the performance of the majority class could appear to be good even though it is misleading (He and Garcia, 2009).

Table 3. Evaluation framework

Component	Method
Primary discrimination	TPR@5%FPR
Ranking measures	PR-AUC and ROC-AUC
Frozen-threshold measures	Achieved FPR, precision, recall, F1, accuracy, balanced accuracy
Model uncertainty	1,000 stratified bootstrap resamples
Pairwise comparison	Paired bootstrap 95% confidence intervals
Explanation	Kernel SHAP
Predictive relevance	Top-five SHAP feature-removal analysis
Agreement	Stack vs Random Forest; Spearman correlation and top-10 overlap
Stability	Seed 42 compared with seeds 7, 21 and 101
Fairness	Predictive equality by age
Robustness	Base-trained model transferred to Variant IV month 7

3.7.1 Statistical Uncertainty

To quantify uncertainty, we relied on 1,000 stratified bootstrap samples of the held-out test set. Legitimate and fraudulent observations were sampled separately and with replacement to maintain the observed class counts. 95% confidence intervals for TPR@5%FPR, PR-AUC and ROC-AUC were computed using percentile intervals. To assess model-to-model differences, we relied on paired bootstrap resampling. The same resampled requests were employed for each competing model, and the metric difference was calculated in each bootstrap iteration. A paired 95% confidence interval that included zero was interpreted to mean that the performance difference, under the resampling framework, was indistinguishable from zero; other intervals were interpreted to mean that the performance difference was distinguishable from zero. No p-values were computed for these comparisons, since several paired comparisons were performed. Consequently, results for which confidence intervals just excluded zero were treated with caution rather than as definitive evidence of large practical superiority.

3.8 Explainability and Explanation Validation

SHAP was employed to understand the deployed stacking ensemble (Lundberg and Lee, 2017). The complete heterogeneous stack cannot be treated as a single tree ensemble for end-to-end TreeSHAP, consequently, Kernel SHAP was employed to explain the complete prediction function. Kernel SHAP

used global explanation analysis on 25 randomly selected training samples and 200 randomly selected test samples, with random seed 42. Global importance was calculated as mean absolute SHAP values across the explanation sample.

The global explanation sample was formed by selecting 200 applications from the held-out test month using a fixed random seed. There was no stratification of the sample by fraud status. Due to the high imbalance in the test population, global SHAP ranking should be considered a general description of the model behaviour for sampled applications and not a ranking of importance for fraud cases. For each case of true positive, false positive, and false negative, an individual explanation has been developed. In place of extreme examples, the application which scored in the middle of all the examples in each of the prediction-outcome groups was selected. Local explanations were produced with $n_{\text{samples}} = 500$.

Kernel SHAP relies upon the assumption of feature independence to construct groups. In such situations, where features of a prediction are correlated, distorted attributions can be the result (Aas, Jullum and Loland, 2021). SHAP values are not to be considered as a causal effect, but rather an attribution to a model prediction. Constructing attributions is not a confirmation of their reliability. Following the opinion of Doshi-Velez and Kim (2017) that interpretability should be evaluated empirically rather than assumed, three additional checks were applied to the explanations developed.

3.8.1 Fidelity-Oriented Feature Removal

The five SHAP features with the highest ranking were removed and the same three-model stacking architecture was retrained without hyperparameter tuning. Performance was then assessed relative to the deployed model. This is considered a fidelity-oriented feature removal or a predictive relevance test rather than a formal proof for SHAP fidelity. Because some of the removed predictors are categorical variables that are represented by several encoded columns, the intervention is a feature-level rather than an encoded-column-level ablation. The design lacks a matched control where an equal number of randomly chosen or low-ranked features are removed. Because of this, the design can only state that the highly ranked predictors likely contain essential predictive information, but cannot fully state that the SHAP ranking itself is optimal.

3.8.2 Cross-Architecture Agreement

We compared the global Kernel SHAP ranking of the stacking ensemble to the independently fitted Random Forest using the same background observations and explanation sample. We evaluated agreement through Spearman rank correlation across the full feature ranking and overlap of the ten highest-ranked features. We interpret strong agreement as meaning that the broad attribution pattern is not unique to one architecture. However, we emphasize that agreement between models does not ensure the correctness of causal explanation.

3.8.3 Stability

The stacking architecture was refitted using random seeds 7, 21 and 101, and the resulting global rankings were compared with the deployed seed-42 model. The same SHAP background and explanation samples were retained. Spearman correlation and top-ten overlap were used to measure stability. Because the explanation sample and background were fixed, this measures how the model is likely to change as a result of new random initializations. It does not measure the Monte Carlo variance of Kernel SHAP.

3.9 Fairness Audit

We evaluated fairness according to the predictive equality framework within the BAF assessment proposed by Jesus et al., (2022). Predictive equality is concerned with the false positive rate across demographic groups and is applicable here because a false positive indicates an applicant who was incorrectly classified as fraudulent. To analyze how well the deployed model performs, we froze the validation threshold to assess 50-and-under and over-50 applicants, corresponding to the age grouping employed with the BAF benchmark. For each group, we calculated sample size, fraud rate, FPR and TPR. Predictive equality was calculated (among other things) to assess whether the false-positive rate was the same, or approximately the same, for the two groups.

When reporting any fairness concerns, we included the group size and group prevalence, since we can't interpret the error differentials without considering group size and prevalence. For lack of confidence intervals, we described the differences in the predictive equality ratios we reported, and gave the estimates for the smaller group with caution. The fairness analysis implemented is constrained to age. Although BAF considers income and employment status to be other protected attributes, they were not independently assessed in the final analysis. Therefore, no conclusion can be drawn to suggest the model has been assessed for fairness across all protected traits.

3.10 Robustness Under Distribution Shift

To tackle robustness, the Base-trained deployed ensemble was put to the test against Variant IV month 7 without model training, hyperparameter tuning, or threshold adjustment. The cross-variant assessment is based on Bhuiyan et al. (2026) who assessed their models on various other variants to understand model behaviour under distribution shift. Two types of robustness are defined. Ranking robustness is assessed via TPR@5%FPR, PR-AUC, and ROC-AUC. Operational robustness is assessed by keeping the Base validation threshold unchanged and seeing how achieving FPR, recall, precision, and F1 change.

This differentiation is important since a model may retain useful ranking ability while a previously chosen decision threshold may no longer fulfill the intended operating behaviour after distribution shift. PR-AUC and precision are interpreted with more caution since both metrics are sensitive to the prevalence of the positive class, and Base month 7 and Variant IV month 7 contain different fraud prevalence. To see if the predictive-equality audit persists among the Variant IV disparities under controlled cross-variant shift, the age-based predictive-equality audit was repeated. Variant IV is thus used to perform a controlled robustness check rather than to provide evidence of external validity on independent real-world banking data.

3.11 Reproducibility

A primary random seed of 42 was used across the main workflow, and seeds 7, 21, and 101 were used for stability testing. Dataset partitions and preprocessing outputs were saved between notebooks so that subsequent modeling and explanation stages used the same prepared data. The fitted models, validation thresholds, and prediction scores were saved and reused for later stages of analysis. The modeling and explainability handoff was confirmed by recomputing predictions for the deployed ensemble and checking if the predictions agreed with the saved prediction scores to the last twelve decimal places.

The software versions which were captured include scikit-learn 1.6.1, XGBoost 3.2.0, LightGBM 4.6.0 and SHAP 0.51.0. The workflow was structured in such a way that the three notebooks were built to run one after the other using the outputs saved at each stage. Measures such as these enhance the reproducibility of the analysis and allow for the results presented in Chapter 4 to be tracked back to the analytical workflow used.

Chapter 4: Results

This chapter covers the empirical outcomes of the framework described in Chapter 3. The results within this chapter are summarized in Sections 4.1 – 4.5. Section 4.1 summarizes the dataset along with the outcomes of the characterization and the preprocessing steps and describes the limitations within the analysis. The response to each of the research questions is relegated to sections 4.2 – 4.4, and a summary of the findings is presented in Section 4.5. The models were created using data gathered over months 0 to 5 in the BAF Base dataset. Month 6 data was used to calibrate the decision thresholds for a 5% FPR (False Positive Rate), which was then used to evaluate the models over data from month 7, which was set aside for this evaluation. Thresholds were fixed prior to evaluation and used in the assessment of the models.

4.1 Dataset Characterization and Preprocessing

The BAF Base dataset has 1,000,000 account-opening applications and 32 columns and has no missing values. Out of the million applications, only 11,029 (1.10%) are fraudulent, while the rest (988,971, or 98.90%) are legitimate. This high level of legitimate applications means that we cannot use accuracy as our analysis metric. Instead, we look at TPR (True Positive Rate) at a fixed 5% FPR (False Positive Rate). While the dataset claims to have no missing values, there are six features that encode missing values via negative sentinels. The features include `intended_balcon_amount` (74.25%), `prev_address_months_count` (71.29%), `bank_months_count` (25.36%), `current_address_months_count` (0.43%), `session_length_in_minutes` (0.20%) and `device_distinct_emails_8w` (0.04%). A verification step was used to distinguish sentinels from real negative values. Five of the features had only one negative value and consequently were treated as missing, and two features (`credit_risk_score` (containing 164 negative values) and `velocity_6h` (containing 44 negative values)) have negative values and consequently were left unchanged. `intended_balcon_amount` was treated as missing since a transfer amount cannot be negative.

Between months 0 and 5, some sentinel-coded patterns of missingness were associated with the prevalence of fraud in a descriptive manner. For `prev_address_months_count`, the prevalence of fraud was recorded as 0.0132 in cases where the value was missing, but was recorded as 0.0028 in cases where the value was recorded, giving a ratio of about 4.73. The comparable ratios were 2.58 for `intended_balcon_amount` and 1.80 for `bank_months_count`. The rest of the missingness indicators had either weak or inverse relationships with the prevalence of fraud. The six pre-specified missingness indicators were all included in the modeling approach, rather than eliminating or including them based on these descriptive relationships with the target. These results only demonstrate associations and

should not be interpreted as causing fraud through missingness.

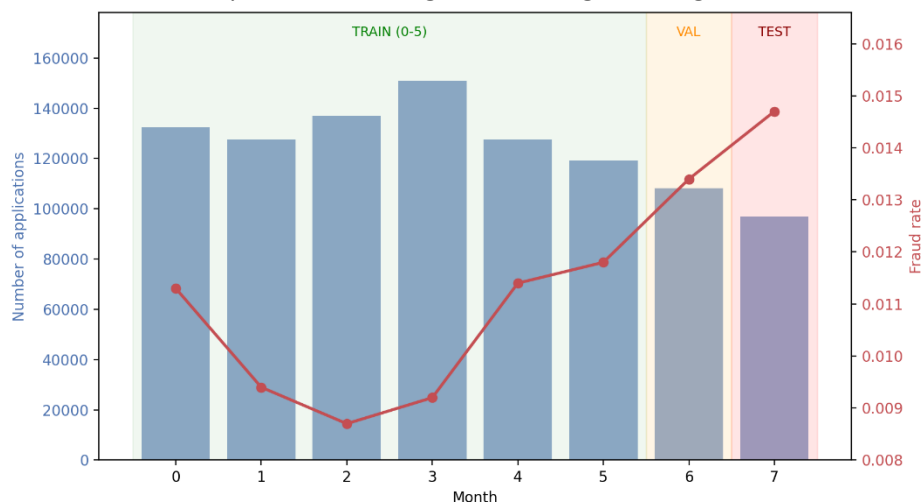


Figure 4: Application volume (bars, left axis) and fraud rate (line, right axis) by month, with train, validation and test boundaries shaded

Since a random split would train on records made after the records used for evaluation, a temporal split was used instead (Table 4). Integrity checks confirmed zero exact duplicate rows, zero duplicate feature vectors, and zero feature vectors that appear in both the training and test months, thus eliminating the leakage due to repetition.

Table 4: Temporal Partitions

Partition	Rows	Share	Fraud cases	Fraud rate
Train (months 0–5)	794,989	79.5%	8,151	1.03%
Validation (month 6)	108,168	10.8%	1,450	1.34%
Test (month 7)	96,843	9.7%	1,428	1.47%

The instances in each partition show a monotonic increase in prevalence, and therefore, all models are trained in conditions that are easier than the conditions where they are evaluated against. The set of predictors consists of 35 features (5 categorical, 30 numeric). We excluded three columns: target, month (a partition artifact), and device_fraud_count which was confirmed to be degenerate in the training data, with a single distinct value and zero variance.

4.2 RQ1: Detection Performance Under Temporal and Distribution Shift

Discrimination in the test month as described in Table 5 is supplemented with 95% stratified bootstrap confidence intervals (1,000 resamples, the classes were resampled separately to retain observed cases); in Figure 5, three metrics that are also described in Table 5 are shown for graphical comparison.

Table 5: Test-month discrimination with 95% bootstrap confidence intervals

Model	TPR@5%FPR	PR-AUC	ROC-AUC
Stacking Ensemble	0.5770 [0.5476, 0.6009]	0.2161 [0.1965, 0.2378]	0.8970 [0.8882, 0.9052]

Model	TPR@5%FPR	PR-AUC	ROC-AUC
Stack + Random Forest	0.5763 [0.5483, 0.6008]	0.2152 [0.1958, 0.2366]	0.8969 [0.8882, 0.9052]
LightGBM	0.5728 [0.5455, 0.5994]	0.2172 [0.1968, 0.2390]	0.8967 [0.8877, 0.9049]
XGBoost	0.5616 [0.5343, 0.5868]	0.2093 [0.1904, 0.2310]	0.8929 [0.8846, 0.9008]
Logistic Regression	0.5336 [0.5056, 0.5623]	0.1849 [0.1684, 0.2044]	0.8869 [0.8777, 0.8956]
Random Forest	0.5182 [0.4909, 0.5462]	0.1689 [0.1518, 0.1886]	0.8771 [0.8671, 0.8861]
Dummy (majority class)	0.0500	0.0147	0.5000

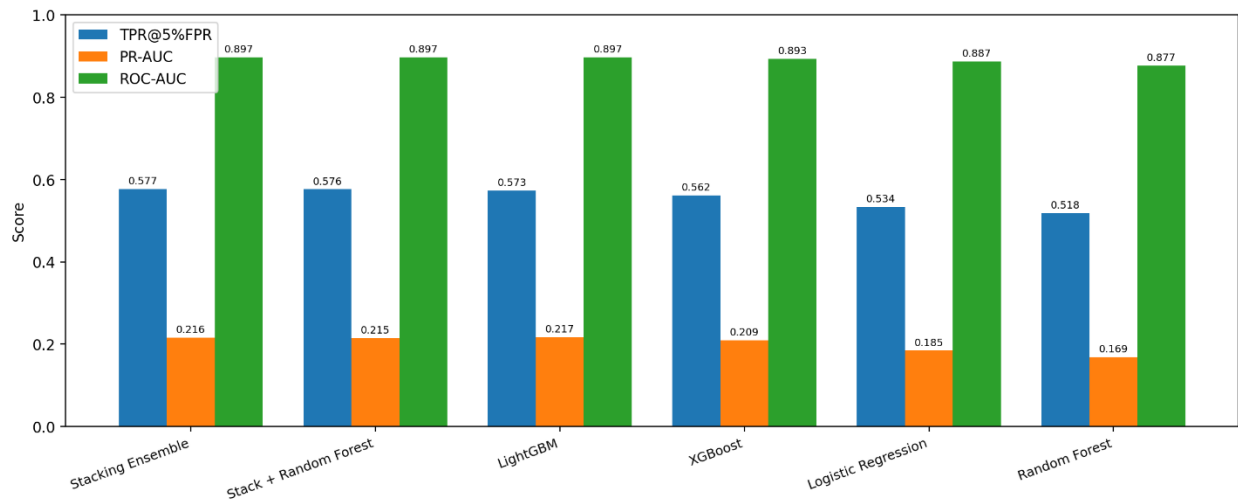


Figure 5: TPR at 5% FPR, PR-AUC and ROC-AUC by model on the test month. The gap between ROC-AUC and PR-AUC across all models illustrates the effect of the 1.47% prevalence.

The empirical foundation for rejecting accuracy at this prevalence is provided by the dummy classifier's 0.9853 accuracy with zero recall. LightGBM had the highest PR-AUC (0.2172), whereas the stacking ensemble had the highest TPR at 5% FPR (0.5770) and ROC-AUC (0.8970). These rankings are formally tested in Section 4.4 because the intervals for the top three models significantly overlap. The whole trade-off surface supporting these summaries is displayed in Figure 6.

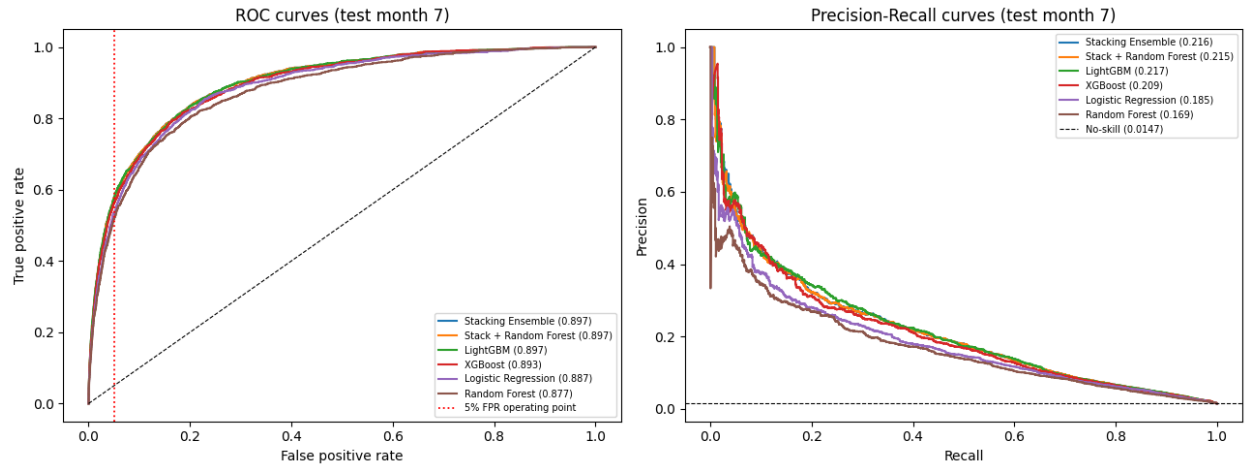


Figure 6: ROC (left) and precision–recall (right) curves on the test month. The dotted vertical line marks the 5% FPR operating point; the dashed horizontal line marks the no-skill precision floor of 0.0147
Figure 7 compares the identical operating-point measurements, whereas Table 6 presents behaviour at each model's frozen threshold.

Table 6: Operating-point metrics at frozen thresholds

Model	Threshold	Achieved FPR	Precision	Recall	F1
Stacking Ensemble	0.8651	0.0391	0.1663	0.5217	0.2522
Stack + Random Forest	0.8658	0.0392	0.1657	0.5196	0.2513
LightGBM	0.8101	0.0426	0.1601	0.5420	0.2472
XGBoost	0.8068	0.0383	0.1660	0.5091	0.2504
Logistic Regression	0.7877	0.0342	0.1646	0.4503	0.2410
Random Forest	0.4813	0.0349	0.1588	0.4405	0.2334

Thresholds calibrated on month 6 at a 5% FPR target and transferred unchanged to month 7. Achieved FPRs differ across models, so this is not a matched-threshold comparison.

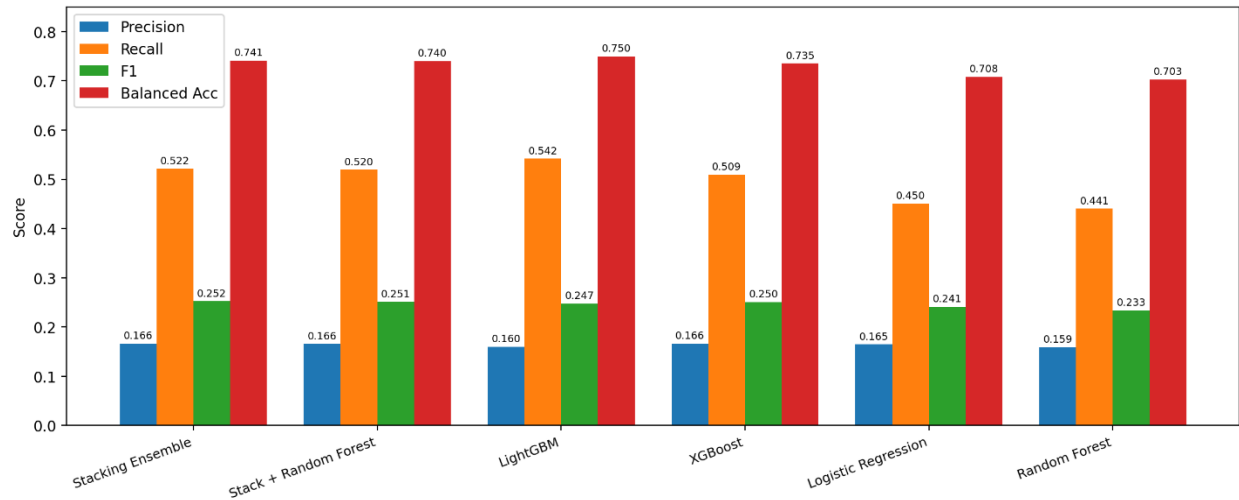


Figure 7: Precision, recall, F1 and balanced accuracy at each model's validation-calibrated frozen threshold.

Of the 4,480 applications that the stacking ensemble identified, 745 were fraudulent and 3,735 were not; 683 of the 1,428 fraudulent cases were overlooked (Figure 8). In comparison to the 5% aim, its realised FPR of 0.0391 shows that the month-6 barrier transferred cautiously rather than precisely. Table 6 is not a matched-threshold comparison because achieved FPRs vary among models (0.0342 to 0.0426). LightGBM's higher recall partially reflects its operating closer to the desired FPR, retrieving 29 more fraudulent applications than the stack at the expense of 325 additional false positives.

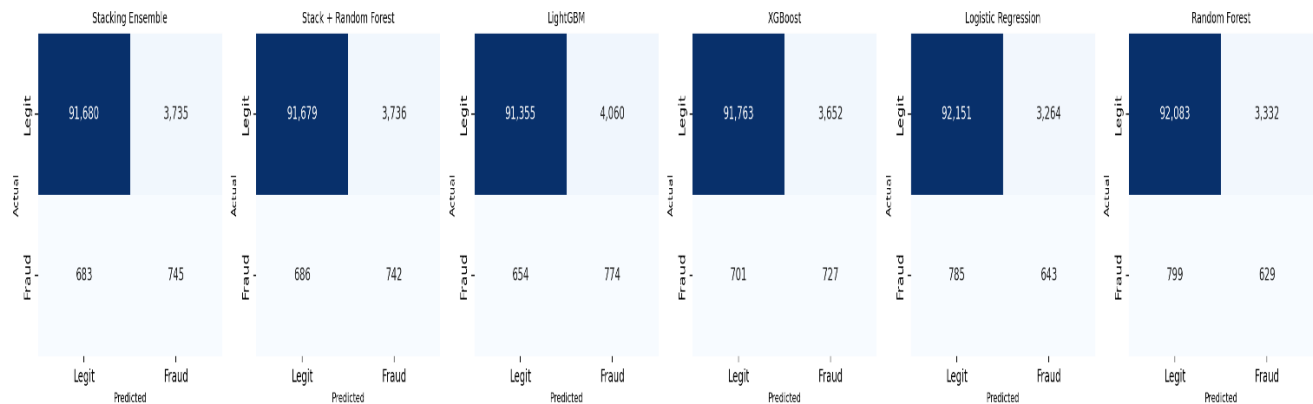


Figure 8: Confusion matrices at frozen thresholds, permitting direct comparison of error composition across models.

The deployed ensemble was applied to month 7 of BAF Variant IV without retraining, retuning, or threshold recalibration in order to evaluate distribution-shift stress test. (Table 7).

Table 7: Robustness under distribution shift

Measure	Base month 7	Variant IV month 7
N	96,843	105,580
Fraud prevalence	1.47%	0.93%
TPR@5%FPR	0.5770	0.4644
PR-AUC	0.2161	0.1058
ROC-AUC	0.8970	0.8681
Frozen-threshold FPR	0.0391	0.0647
Frozen-threshold recall	0.5217	0.5264
Frozen-threshold precision	0.1663	0.0711
Frozen-threshold F1	0.2522	0.1253

Deployed stacking ensemble applied to Variant IV without retraining, retuning or threshold recalibration.

TPR at 5% FPR dropped by 0.1126 (19.5% relative), PR-AUC by 51.0%, and ROC-AUC by 3.2% as ranking performance declined. The decreased occurrence of Variant IV, to which PR-AUC is sensitive by construction, is reflected in the discrepancy between the ROC-AUC and PR-AUC reductions. Operationally, recall remained nearly constant (0.5217 to 0.5264), but precision decreased by 57.2% and realised FPR increased from 0.0391 to 0.0647, above the 5% design target. The model continued to identify a comparable percentage of fraud, but it did so at the cost of significantly more false alarms. These findings suggest significant but insufficient resilience, and threshold transferability is a failure mode different from discriminate loss.

4.3 RQ2: Explainability and Reliability

4.3.1 Global and Local SHAP Attributions

Using 200 held-out test applications and 25 background observations from the training set, Kernel SHAP was applied to the entire deployed stacking ensemble instead of any constituent learner (Figure 9).

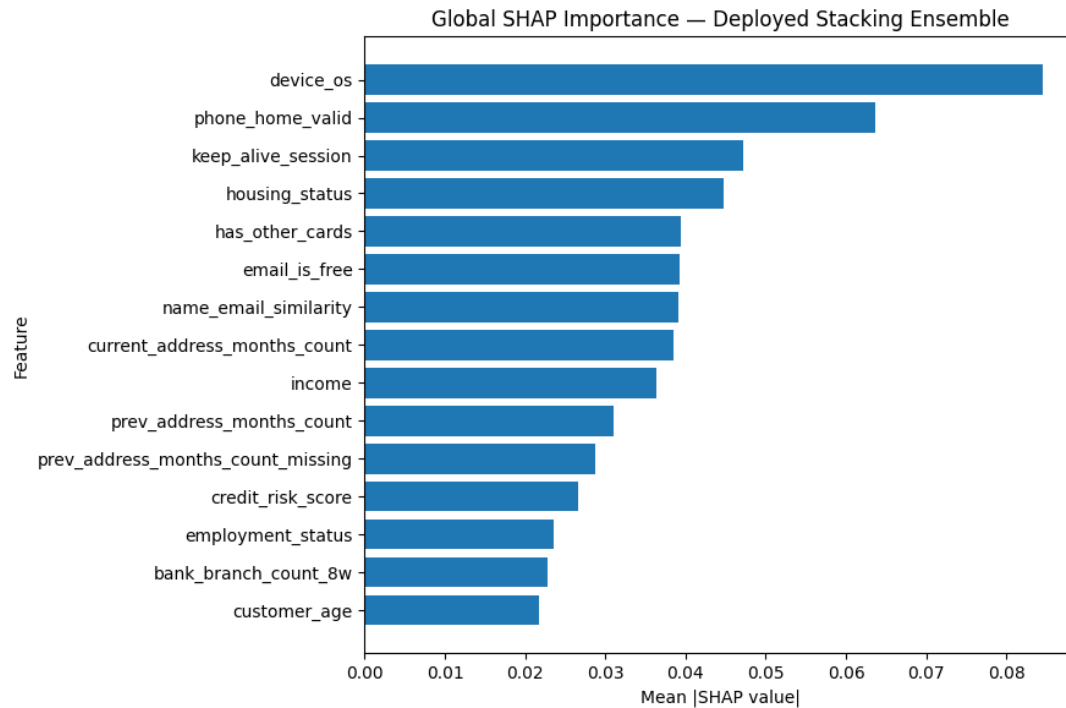


Figure 9: Average absolute SHAP values for the top fifteen ranked features of the implemented stacking ensemble.

The most significant mean absolute SHAP values were device_os (0.0846), phone_home_valid (0.0637), keep_alive_session (0.0472), housing_status (0.0448), has_other_cards (0.0395), email_is_free (0.0392), name_email_similarity (0.0391), current_address_months_count (0.0386), income (0.0364), and prev_address_months_count (0.0311). The count of months with a previous location that are missing was ranked eleventh (0.0287), indicating that the intentional absence of data has an independent effect from the observed value. Figure 10 illustrates the orientation of these impacts.

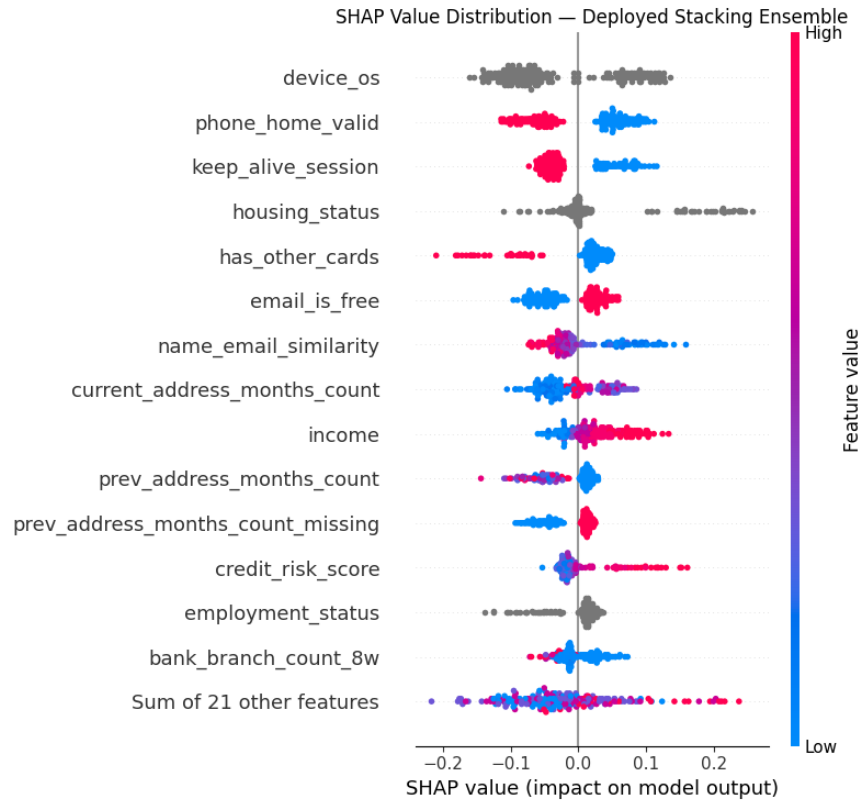


Figure 10: Distribution of SHAP values within the explanatory sample. For categorical attributes, the colour gradient lacks any sequential high–low significance

Three median-scoring instances were chosen for local elucidation at the fixed threshold of 0.8651: a true positive (score 0.9241), a false positive (0.9060), and a false negative (0.6002), illustrated in Figure 11. In each instance, the result stemmed from a confluence of characteristics rather than a singular preeminent factor, and the identical factors manifested with contrasting signals in different circumstances. In the case of the false positive, the affirmative impacts of housing_status, device_os, and name_email_similarity surpassed the detrimental effects of phone_home_valid and keep_alive_session; conversely, in the false negative, the same variables had an opposing outcome.

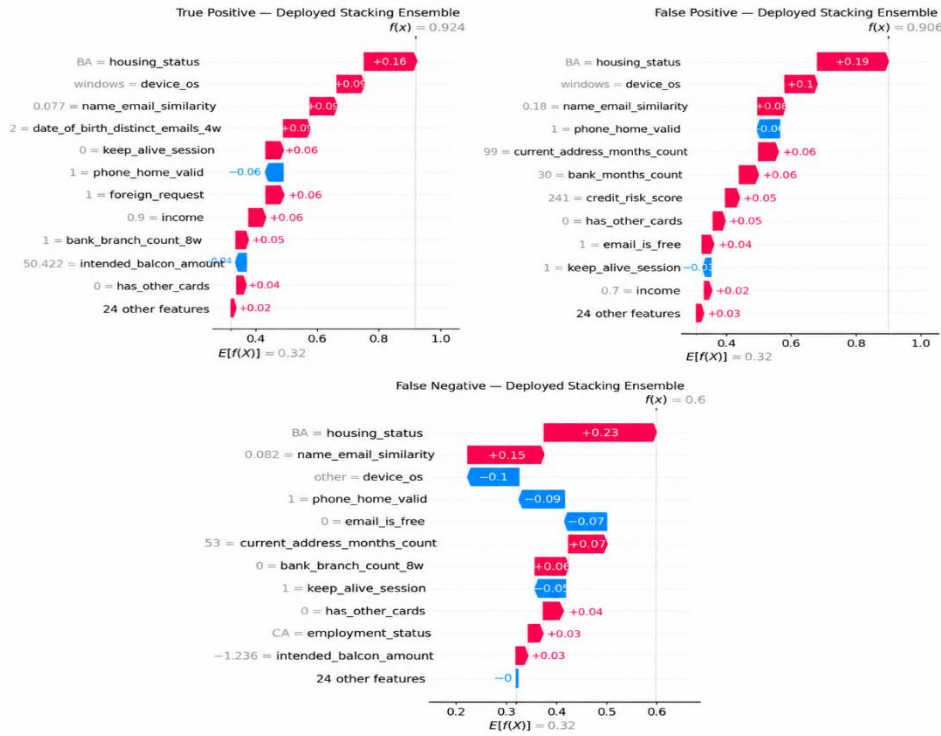


Figure 11: SHAP waterfall explanations for (a) a true positive, (b) a false positive and (c) a false negative. The baseline $E[f(X)]$ is the expected model output over the SHAP background sample, not the dataset fraud prevalence.

4.3.2 Explanation Reliability

Generating attributions does not confirm their reliability, hence three supplementary assessments were conducted (Table 8).

Table 8: Explanation reliability tests

Test	Result
Fidelity: TPR@5%FPR after removing top-5 SHAP features	0.5770 $\rightarrow$ 0.4314 (-0.1457, 25.2%)
Fidelity: PR-AUC after removing top-5 SHAP features	0.2161 $\rightarrow$ 0.1444 (-0.0717, 33.2%)
Agreement with Random Forest (Spearman ρ , 35 features)	0.9633 ($p = 2.02 \times 10^{-20}$)
Agreement: top-10 feature overlap	9/10
Stability across seeds 7, 21, 101 (Spearman ρ)	0.9986, 0.9969, 0.9969
Stability: top-10 overlap	10/10 for all three seeds

Eliminating the five top-ranked SHAP features and retraining the static architecture without adjustments diminished TPR@5%FPR from 0.5770 to 0.4314 and PR-AUC from 0.2161 to 0.1444. This demonstrates that these characteristics provide significant predictive value. Nonetheless, due to the absence of a

matched random or low-ranked feature removal control, the experiment should not be construed as definitive evidence of SHAP faithfulness or the superiority of the SHAP ranking.

4.3.3 Fairness Audit

Predictive equality was evaluated at the frozen threshold by the age categorisation of the BAF benchmark (Table 9).

Table 9: Predictive equality by customer age

Dataset	Group	N	Prevalence	FPR	TPR
Base month 7	Age ≤ 50	93,901	1.35%	0.0365	0.5000
Base month 7	Age > 50	2,942	5.51%	0.1266	0.6914
Variant IV	Age ≤ 50	92,366	0.87%	0.0550	0.5000
Variant IV	Age > 50	13,214	1.39%	0.1326	0.6413

The predictive-equality ratio stood at 0.2884 in Base month 7, accompanied by an absolute FPR disparity of 0.0901. In month 7 of Variant IV, the predictive-equality ratio enhanced to 0.4150, while the absolute FPR gap diminished to 0.0776.

During the Base test month, those over the age of 50 had a false-positive rate of 0.1266, in contrast to a rate of 0.0365 for those aged 50 or younger. This resulted in a predictive-equality ratio of 0.2884 and an absolute false positive rate gap of 0.0901. The senior category exhibited a greater true-positive rate (0.6914 compared to 0.5000) and a significantly elevated incidence of fraud (5.51% versus 1.35%). In Variant IV, the rates of false positives were 0.1326 for candidates older than 50 and 0.0550 for those aged 50 or younger, resulting in a predictive-equality ratio of 0.4150 and an absolute disparity of 0.0776. Despite enhancements in prediction equality throughout the cross-variant shift, a significant subgroup disparity persisted. These numbers illustrate the discrepancies in mistake rates found and do not, on their own, confirm discrimination or a causal relationship with age.

4.4 RQ3: Is Additional Model Complexity Justified?

The paired bootstrap comparison evaluated if the disparities surpass sampling variability. Every pair was assessed using the same resamples, maintaining the correlation established by evaluating rival models on the identical applications (Table 10).

Table 10: Paired bootstrap comparisons (95% CI on the difference)

Comparison	Metric	Difference	95% CI	Distinguishable
Stack vs LightGBM	TPR@5%FPR	+0.0042	[−0.0063, +0.0140]	No
Stack vs LightGBM	PR-AUC	−0.0011	[−0.0063, +0.0037]	No
Stack vs XGBoost	TPR@5%FPR	+0.0154	[+0.0049, +0.0259]	Yes
Stack vs XGBoost	PR-AUC	+0.0068	[+0.0014, +0.0124]	Yes
Stack vs Stack + RF	TPR@5%FPR	+0.0007	[−0.0014, +0.0028]	No
Stack vs Stack + RF	PR-AUC	+0.0009	[+0.0002, +0.0017]	Yes
LightGBM vs Logistic Regression	TPR@5%FPR	+0.0392	[+0.0210, +0.0588]	Yes

Comparison	Metric	Difference	95% CI	Distinguishable
LightGBM vs Logistic Regression	PR-AUC	+0.0323	[+0.0199, +0.0451]	Yes
XGBoost vs Logistic Regression	TPR@5%FPR	+0.0280	[+0.0091, +0.0462]	Yes
XGBoost vs Logistic Regression	PR-AUC	+0.0244	[+0.0124, +0.0370]	Yes

Each pair evaluated on identical bootstrap resamples. An interval excluding zero indicates a difference distinguishable from sampling variation.

The move from a linear baseline to gradient boosting is clearly supported: all four intervals comparing LightGBM and XGBoost against Logistic Regression exclude zero. The case for stacking is more qualified. The ensemble significantly outperformed XGBoost on both metrics, but its 0.0042 advantage over LightGBM on the primary metric was not distinguishable from zero, and on PR-AUC LightGBM held a marginally higher point estimate. Adding a fourth base learner produced no distinguishable gain in TPR at 5% FPR; the PR-AUC difference of 0.0009 favoured the simpler three-learner stack, while the four-learner variant required 972.3 seconds to train against 222.8 seconds, a 4.4-fold increase.

Architecture diagnostics explain this pattern (Table 11). All three base learners received positive meta-learner coefficients, but their predictions were highly correlated. Limited predictive diversity offers a direct explanation for why stacking yields only marginal gains over its strongest constituent.

Table 11: Meta-learner coefficients and base-learner prediction correlations

Base learner	Meta coefficient	Share of absolute magnitude	Correlation with XGBoost	with LightGBM
XGBoost	+1.4732	25.9%	—	0.9771
LightGBM	+3.1127	54.6%	0.9771	—
Logistic Regression	+1.1127	19.5%	0.9111	0.9218

Meta-learner intercept: -2.6798 . Coefficients are interpreted descriptively rather than as literal percentage contributions to individual predictions.

4.5 Summary of results

The dataset is significantly skewed (1.10% fraud), demonstrates temporal fluctuations in fraud occurrence, and includes six fields regarded as sentinel-coded missing values. During the descriptive analysis of the training window (months 0–5), the `prev_address_months_count` exhibited the highest ratio of missing to present fraud prevalence among the six variables, indicating that fraud prevalence was roughly 4.73 times greater when the value was absent compared to when it was available.

Addressing RQ1, the stacking ensemble obtained a metric of TPR@5%FPR equal to 0.5770 on the held-out month-7 test set. Furthermore, the model achieved recall = 0.5217 at FPR = 0.0391 when the threshold calibrated on validation month 6 was moved to month 7 without any changes. It was able to detect 745 of the 1,428 fraud cases, and issued 3,735 false positives, and missed 683 fraud cases. With the transition to Variant IV, the ranking efficacy diminished (TPR 0.4644; PR-AUC 0.1058), and the applied threshold resulted in an FPR of 0.0647, compared to a 5% design objective, effectively reducing precision by fifty percent. Robustness is consequently limited, and threshold recalibration arises as a necessary operational procedure apart from retraining.

In relation to RQ2, SHAP attributions utilised across the complete ensemble recognised device, contact-validation, session, and address-history features as predominant, with both the presence and lack of previous address history influencing independently. The attributions passed all three reliability assessments. Predictive-equality ratios for Base and Variant IV were 0.2884 and 0.4150, respectively, according to the fairness audit. Applicants over 50 still had a much greater false-positive rate than those under 50, despite improvements in cross-variant predictive equivalence.

In relation to RQ3, the three-learner stacking ensemble gave the best point estimate for the primary TPR@5%FPR metric (0.5770) and achieved the best ROC-AUC (0.8970), while LightGBM achieved the best PR-AUC (0.2172 as compared to the stack's 0.2161). The stack and LightGBM had a TPR@5%FPR paired bootstrap difference of +0.0042 with a 95% CI of [-0.0063, +0.0140] and -0.0011 with a 95% CI of [-0.0063, 0.0037], respectively. Since both intervals included zero, this analysis is unable to state that the stack is superior to LightGBM for these metrics. The stack also was superior to XGBoost for both TPR@5%FPR (result: +0.0154, 95% CI: +0.0049, +0.0259) and PR-AUC (result: +0.0068, 95% CI: +0.0014, +0.0124), both of which had paired intervals that excluded zero. For the given frozen threshold, the stack issued 354 fewer alerts than LightGBM, but detected 29 fewer fraud cases. As to whether this is a good operational trade-off depends upon the relative costs of undetected fraud and spare capacity for manual reviews. The four-base-learner Stack + Random Forest ablation did not improve the primary TPR point estimate and increased training cost, justifying the added complexity little if at all.

CHAPTER 5: DISCUSSION

5.1 Introduction

This chapter provides an answer to the questions posed in the introduction and in Chapter 2. This includes fraud detection in a situation of high class imbalance and temporal changes. It also includes the reliability and usefulness of SHAP explanations, subgroup fairness, and whether the additional complexity caused by stacking results in an improvement in performance over simpler methods. The data reveals that predictive performance, explainability, stability, and fairness should be analyzed collectively.

5.2 Discussion of RQ1: Detection Performance Under Temporal and Distribution Shift

The aim of RQ1 was to evaluate the stacking ensemble's performance at detecting identity fraud at the opening of bank accounts in cases of temporal and distribution shifts. The results show the inadequacy of using accuracy as a means of assessment for this type of detection problem. An accuracy of 0.9853 was achieved by the majority-class Dummy Classifier when no fraudulent applications were detected. This is a common issue in highly imbalanced classification, in which a model reaches high levels of accuracy by classifying the majority class (He and Garcia, 2009; Branco, Torgo and Ribeiro, 2016). As such, the evaluation of fraud detection, in accordance with Saito and Rehmsmeier (2015) and the evaluation framework of Jesus et al. (2022), will focus on PR-AUC and TPR at a fixed 5% of FPR.

At the end of month 7, the stacking ensemble obtained the best TPR@5%FPR estimate of 0.5770 and had a value of the ROC-AUC of 0.8970, the best among all models, while the best value of the PR-AUC was obtained by LightGBM, with a score of 0.2172, while the stack scored 0.2161. Because operating metrics show different trends and depend on the operational needs, preference for a certain model may

differ. This supports the findings of Sun et al. (2025) who claim that the architecture of the model is not the only factor to consider for dealing with extreme cases of class imbalance. It also indicates the choice of operating points and optimization can affect the results and outcomes of fraud detection.

One important outcome is the difference between the stacking ensemble TPR@5%FPR of 0.5770 and the score for recall, which was equal to 0.5217. During model validation, the threshold was fixed at the value of 0.8651, and the actual value of the FPR was 0.0391 instead of 0.05. This demonstrates that a threshold decided on one month may not guarantee a consistent operating behaviour in subsequent months. The model identified 745 fraudulent cases and generated a total of 3,735 false positives while missing 683 fraudulent cases. This clearly shows temporal assessment of fraud detection models is very important, as changing behavioural patterns in a population can also influence the rank of a model used for fraud detection.

The problem is further illustrated by Variant IV analysis. There was no retraining or threshold adjustment when the Base-trained ensemble was transferred. In this case, TPR@5%FPR worsened from 0.5770 to 0.4644, while ROC-AUC deteriorated from 0.8970 to 0.8681. PR-AUC fell from 0.2161 to 0.1058. We must consider that fraud prevalence dropped in Variant IV; as such, we should be careful to interpret PR-AUC. Of greater concern, the frozen-threshold FPR worsened from 0.0391 to 0.0647 and exceeded the 5% threshold, and the corresponding precision fell to 0.0711.

Thus, some ability to detect fraud was preserved under distribution shift, but its operational behaviour was worse. We therefore, answer RQ1 positively but with caveats, as the ensemble demonstrated some predictive ability in the evaluation performed in the future, along the time axis, even though we cannot assume that either its performance or its decision threshold will remain stable across different distributions. In practice, a Fraud Detection System would need continual monitoring and threshold adjustment as opposed to a one-time deployment.

5.3 Discussion of RQ2: Explainability, Reliability and Fairness

RQ2 evaluated if SHAP explanations for the deployed ensemble were sufficient, as well as if the explanations were reliable to support examination of individual instances. In terms of SHAP importance ranking, across all ensembles, the most relevant features were `device_os`, `phone_home_valid`, `keep_alive_session`, `housing_status`, and `has_other_cards`. Other address-history features retained importance. This included `prev_address_months_count` and its missing value indicator. The address-history features remaining important corresponds to what we observed in the descriptive statistics. In particular, `prev_address_months_count` was missing for fraud cases, which happened 4.73 times more than when the value was not missing. This further substantiates that the patterns we identified in the data when explaining the model ensemble remained in the final deployed model. That said, we need to not consider the associations as causal. Shapley values explain how the features in the model contribute to the prediction function. It is not meant to illustrate that the features cause fraud (Lundberg and Lee, 2017).

In terms of individual classifications, SHAP explanations showed that the classifications were not driven by a single dominant feature. In the false positives example, the legitimate applicant received a classification score of 0.9060, which was determined by the combination of multiple positive features and negative protective features. In the false negatives example, the classification score was 0.6002 because the risk-increasing features were not sufficient to overcome the protective negative features.

The SHAP explanations localize the reason why the application was above or below the decision threshold for examiner review. This example exemplifies the importance of local explanations in Zhou et al (2023) who emphasize the importance of explanations in financial fraud-detection that is geared towards the user.

Producing SHAP plots alone will not show how well explanation can be generalized. Therefore, the study examined three other characteristics. With five of the top SHAP features removed, TPR@5%FPR was reduced from 0.5770 to 0.4314, PR-AUC from 0.2161 to 0.1444, suggesting that the top features likely contain predictive information. In addition, there was considerable agreement with Random Forest, with Spearman $\rho = 0.9633$, and 9 of the top 10 features in common. Finally, when 3 alternative seeds were used to retrain the stacking architecture, the Spearman correlations were 0.9969 and 0.9986, and top 10 agreements were maintained. The results of the study support the case that the main global SHAP ranking is reproducible and provide evidence of predictive relevance. However, it must be said that the SHAP ranking is evidence of little else than predictive relevance. Kumar et al. (2020) state that Shapley value explanations should be used with caution, as they may lead to overestimation of feature importance. Aas, Jullum and Loland (2021) argue that correlations among predictors influence the SHAP attribution. In addition, the feature removal experiment did not contain a control where randomly chosen or low ranked features were removed, and therefore was only able to show that the features that were removed have predictive value. Therefore, the SHAP ranking should not be viewed as optimal.

Additionally, care must be taken with the fairness analysis. The predictive-equality ratio was 0.2884 for the Base test month, with a false-positive rate of 0.0365 for applicants under 50 and 0.1266 for those over 50. Predictive equality increased to 0.4150 under Variant IV, with corresponding false-positive rates of 0.0550 and 0.1326. As a result, even if the disparity's relative magnitude decreased, it remained under the cross-variant shift. These results also align with the broader fairness issues documented by Jesus et al., (2022), where excellent predictive performance does not guarantee equitable error rates across the protected groups. Although customer age scored roughly sixteenth in the worldwide ranking, it was not one of the highest-ranked SHAP variables; yet, significant subgroup disparities in false-positive rates persisted.

RQ2 is answered with caution. SHAP includes useful global and individual transparency, and the overall attribution pattern showed stability and predictive relevance. Still, the results should not be used to make judgements on adverse effect. As Rudin (2019) states, post-hoc explanations in high-stakes situations should not be employed as substitutes for human judgment and should instead complement it.

5.4 Discussion of RQ3: Is Additional Model Complexity Justified?

RQ3 assessed whether greater model complexity was justified in terms of greater model performance. There was a clear distinction between the architectural changes from Logistic Regression to Gradient Boosting and from Gradient Boosting to Stacking.

Logistic Regression was significantly outperformed by all of XGBoost, LightGBM, and Stack. LightGBM outperformed the linear model by 0.0392 in TPR@5%FPR and PR-AUC by 0.0323. XGBoost outperformed it by 0.0280 and 0.0244 respectively. All confidence intervals were fully above zero. This shows that non-linear Gradient Boosted models had predictive elements that the linear model lacked.

The evidence for stacking over LightGBM was much weaker. Stack outperformed LightGBM in TPR@5%FPR by 0.0042, with a 95% CI of $[-0.0063, 0.0140]$, and its PR-AUC was also slightly lower. Therefore, it was not distinguishable from LightGBM on either measure. However, Stack did significantly improve on XGBoost.

The architecture diagnostics help explain the limited benefit. Prediction correlations between base learners ranged from 0.9111 to 0.9771, with XGBoost and LightGBM particularly highly correlated. Stacking generalization relies on complementary information from component models (Wolpert, 1992). When predictions are almost the same, very little additional information is available to the meta-learner.

The Random-Forest ablation supports this interpretation. Adding a fourth learner showed an increase in training time (from 222.8 to 972.3 seconds) and did not positively impact TPR@5%FPR. More complex architecture did not improve performance.

RQ3 therefore confirms that moving from Logistic Regression to gradient boosting is justified, while the additional move from LightGBM to stacking provides only limited additional value. This supports Rudin's (2019) position that additional opacity in high-stakes models should be justified by evidentiary benefit, not considered best practice.

5.5 Practical Conclusions and Limitations

The results show that fraud-detection systems should be looked at as systems that should not be left as static models, but as systems that are constantly being monitored. Important to this point is considering the calibration of thresholds. Variant IV indicates that an acceptable threshold can, after a shift in the data distribution, bring about many false positives.

The study also confirms that explainability, fairness, and performance should be analyzed individually. Rankings of SHAP remained the same after stabilization and still did not eliminate error gaps with respect to age. The most complex of the models also did not result in significant improvements compared with the strongest of the individual boosted learners. It can therefore be expected that in practice SHAP explanations should be intended for analysts for review and model governance and should not provide automated explanations for rejecting applicants, in line with advice provided by both The Information Commissioner's Office and The Alan Turing Institute (2022), and the risk management principles of Tabassi (2023).

This research is constrained by the artificial characteristics of BAF, the application of non-temporal stratified cv=3 for internal stacking, the limited Kernel SHAP background and explanation samples, and a fairness assessment restricted to age. Other studies should therefore consider the use of independent, real-world data, examine out of fold stacking in a temporal order, try other samples of SHAP background, controlled feature removal, and a wider range of protected characteristics.

6. Conclusion

6.1 Summary of the Study

This dissertation has developed and critically appraised an explainable AI framework to detect identity fraud in account opening using the BAF NeurIPS 2022 dataset. The proposed ensemble uses stacking to combine the three classifiers XGBoost, LightGBM and Logistic Regression. Performance of the framework was evaluated using metrics suitable to handle severe class imbalance such as PR-AUC and TPR at a false-positive rate (FPR) of 5%. The study incorporated temporal partitioning, bootstrap uncertainty, SHAP explanations, reliability testing, fairness, and controlled robustness testing in Variant IV.

6.2 Conclusions Related to the Research Questions

For the first research question, in the held-out temporal evaluation, the stacking ensemble showed good fraud detection capability. The stacking model had the best TPR at 5% FPR being 0.5770 and the best ROC-AUC at 0.8970, and was just slightly behind LightGBM in PR-AUC. For the validation-calibrated threshold, the model achieved a recall of 0.5217 at an observed FPR of 0.0391. In Variant IV, the TPR at 5% FPR fell to 0.4644 and the frozen threshold FPR increased to 0.0647, indicating the model retained some degree of predictive capability, but demonstrated threshold fragility to distribution shifts.

In relation to RQ2, SHAP provided both global and individual explanations of the deployed ensemble. According to the ensemble, the leading features were `device_os`, `phone_home_valid`, `keep_alive_session`, `housing_status`, and `has_other_cards`. Three supporting correlations validated the explanations: removing the top 5 SHAP features significantly reduced performance; strong agreement with Random Forest was observed (Spearman's $\rho = 0.9633$); and feature importance remained consistent in all random seeds. Despite an improvement under the cross-variant shift, the fairness audit yielded predictive-equality ratios of 0.2884 on Base and 0.4150 on Variant IV, showing significant subgroup variations in false-positive rates. SHAP should be used to aid human review of explanations for decisions and not be used as a justification for an adverse decision.

For RQ3, the stacking ensemble produced the best point estimate of the primary measure and significantly outperformed XGBoost on TPR@5%FPR and PR-AUC. The difference between LightGBM and the stacking ensemble was also small, demonstrating that the most significant gains from complexity were achieved when the interpretable, linear baseline was exceeded to gradient boosting. The high correlations among the base learners (0.9111 to 0.9771) explain close ensemble performance and illustrate diminishing returns from additional models and suggest that future improvements should emphasize more diverse learners.

6.3 Contribution and Final Conclusion

The main contribution of this work is the development and integration of heterogeneous stacking, fixed-operating-point analysis, SHAP-based explanation, reliability of explanations, fairness checks, and cross-variant robustness in a single bank account opening fraud detection system. This work argues that these aspects should be analyzed cumulatively, rather than thinking that an increase in the complexity of prediction will bring a sufficient advantage in practice.

In terms of performance, the stacking ensemble produced the highest estimated values for TPR@5%FPR and ROC-AUC, but in the paired bootstrap analysis, the performance of the LightGBM model on TPR@5%FPR and PR-AUC was indistinguishable. Therefore, there is not sufficient evidence to prefer the more complex stacking over its best individual component.

In addition, this work shows that the assessment of account opening fraud models requires more than the performance metrics of predictive models. Reliability of explanations and subgroup fairness and performance under distribution shift should be analyzed. Future work will analyze the use of diverse base learners, temporally ordered out-of-fold stacking, a broader evaluation of fairness checks, and the use of real-world banking data.

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Appendices

Appendix A - Data Collection

Table A1. Data source and collection summary

Item	Description
Research data type	Secondary, publicly available synthetic data; no primary participant data were collected.
Dataset	Bank Account Fraud (BAF) NeurIPS 2022 dataset developed by Jesus et al. (2022).
Dataset files used	Base and Variant IV.
Data format	CSV files.
Number of records	1,000,000 applications in Base and 1,000,000 applications in Variant IV.
Original variables	32 columns: 30 predictive variables, fraud_bool as the binary target, and month as the temporal variable.
Temporal coverage	Eight monthly periods, coded from month 0 to month 7.
Primary development dataset	Base dataset.
Robustness dataset	Variant IV, used for controlled cross-variant robustness evaluation and not for model training, hyperparameter tuning or threshold selection.
Target variable	fraud_bool, where the positive class represents fraudulent account-opening applications.
Base fraud prevalence	11,029 fraudulent applications out of 1,000,000 (approximately 1.10%).
Training partition	Months 0-5: 794,989 applications and 8,151 fraud cases.
Validation partition	Month 6: 108,168 applications and 1,450 fraud cases.
Test partition	Month 7: 96,843 applications and 1,428 fraud cases.
Ethical/privacy consideration	The BAF data are synthetic and publicly available; no identifiable banking-customer data were processed.

Appendix B- Dataset Preparation and Feature Configuration

Area	Item	Result / configuration
Dataset profile	Base dataset	1,000,000 applications; 32 original columns; 11,029 fraud cases (1.10%).
Sentinel missingness	intended_balcon_amount	742,523 negative-coded records (74.25%).
Sentinel missingness	prev_address_months_count	712,920 negative-coded records (71.29%).
Sentinel missingness	bank_months_count	253,635 negative-coded records (25.36%).
Sentinel missingness	current_address_months_count	4,254 negative-coded records (0.43%).
Sentinel missingness	session_length_in_minutes	2,015 negative-coded records (0.20%).
Sentinel missingness	device_distinct_emails_8w	359 negative-coded records (0.04%).
Training-only missingness analysis	prev_address_months_count	Fraud prevalence 0.0132 when missing vs 0.0028 when present; ratio 4.73.
Training-only missingness analysis	intended_balcon_amount	0.0122 vs 0.0047; ratio 2.58.
Training-only missingness analysis	bank_months_count	0.0154 vs 0.0085; ratio 1.80.
Temporal partition	Training	Months 0-5; 794,989 applications; 8,151 fraud cases; 1.03% fraud.
Temporal partition	Validation	Month 6; 108,168 applications; 1,450 fraud cases; 1.34% fraud.
Temporal partition	Test	Month 7; 96,843 applications; 1,428 fraud cases; 1.47% fraud.
Final feature set	Predictors	35 predictors: 5 categorical and 30 numerical; six missingness indicators added.
Excluded fields	fraud_bool / month / device_fraud_count	Target excluded; month used only for temporal splitting; device_fraud_count excluded as degenerate in training.
Integrity checks	Duplicates / train-test overlap	0 exact duplicate rows; 0 duplicate predictor vectors; 0 predictor-vector overlap between train months 0-5 and test month 7.

Appendix C - Data Preparation Pipeline

Table C1. Data preparation and preprocessing pipeline

Stage	Procedure implemented	Output/Purpose
1. Data loading	Base and Variant IV CSV files were loaded into Python using pandas.	Created working data frames for subsequent analysis.
2. Schema verification	The two datasets were checked to confirm that they shared the same feature structure.	Ensured the same preprocessing procedure could later be applied to Variant IV.
3. Class-distribution assessment	fraud_bool prevalence was examined across Base and across monthly periods.	Confirmed severe class imbalance and temporal variation in fraud prevalence.
4. Missing-value audit	Six documented variables containing sentinel-coded missing information were identified.	Distinguished encoded missingness from conventional null values.
5. Sentinel verification	Negative-valued features were examined individually. Five repeated -1 fields were treated as sentinel-coded missingness, while genuine negative measurements such as credit_risk_score and velocity_6h were retained.	Prevented valid negative values from being incorrectly treated as missing.
6. intended_balcon_amount handling	Negative values were treated as missing according to the BAF data definition rather than using only the -1 rule.	Accounted for the feature's different missing-value encoding.
7. Missingness indicators	A binary missingness indicator was created for each of the six sentinel-coded fields while retaining the original values.	Preserved both original feature information and missingness patterns.
8. Duplicate/leakage checks	Exact duplicates, duplicate predictor vectors and train-test predictor-vector overlap were checked.	Reduced the possibility of inflated performance from repeated observations.
9. Feature exclusion	fraud_bool was excluded from predictors; month was reserved for temporal partitioning; device_fraud_count was removed because it was constant in training.	Prevented target leakage and removed a non-informative predictor.

10. Final feature set	Six missingness indicators were added to the remaining predictors.	Produced 35 final predictors: 5 categorical and 30 numerical variables.
11. Categorical encoding	One-hot encoding was applied to categorical variables, with unknown categories ignored at prediction time.	Converted categorical features into model-compatible numerical form.
12. Numerical preprocessing	Numerical variables were unchanged for tree models; StandardScaler was applied within the Logistic Regression pipeline.	Applied model-appropriate preprocessing.
13. Temporal partitioning	Months 0-5 were used for training, month 6 for validation/threshold calibration, and month 7 for terminal testing.	Preserved the temporal evaluation design and isolated the final test month.
14. Saved outputs	Prepared train, validation and test partitions and the feature configuration were saved for use by later notebooks.	Ensured consistent hand-off from preparation to modelling and explainability.